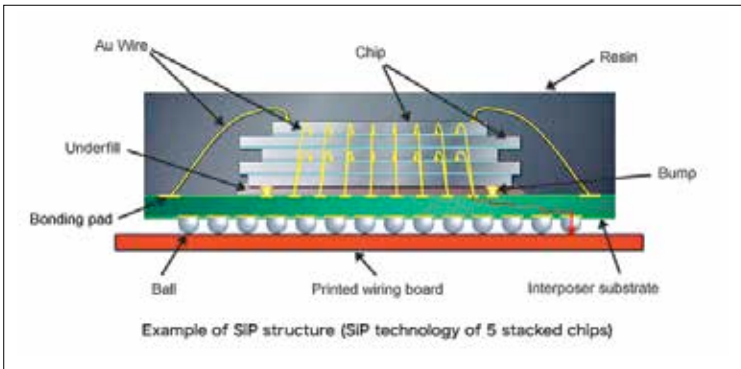


the IC has been etched. In fabrication terminology, a large thin sheet of substrate on which multiple ICs can be drawn is called a wafer. The terms substrate and wafer are sometimes used interchangeably.

5. **Package:** A circuit built on a die is often very fine and can have microscopic elements which can be easily damaged. In addition, the components of the circuit can also get destroyed by moisture or by reacting with other elements present in the ambient air. To protect against these, the die must be put inside a protective casing which is airtight, covers the die on all sides and is a good thermal conductor. Such a casing is called a package. Normally, when we pick up an IC, we pick up the package. Inside the (typically) black cuboid lies the real circuitry (the die). The pins coming out of the package are connected to the die while packaging.



SiP is MCM stacked vertically

6. **Passive Component:** Passive components on a system are the elements which do not change their behaviour based on another electric signal. Capacitors, Resistors, Sensors, LEDs, Speakers would qualify as passive devices.

That was the basic terminology related to SoC, SiP, SoP and MCM. If you are wondering what these are, read on.

1. **SoC:** A system-on-a-chip – one package contains multiple units which do different tasks. You've read about this a few times by now.
2. **MCM:** MCM or Multi-Chip Module is a name which sometimes predates SoCs but has a very similar function. MCMs too embed multiple circuits that can do different jobs on the same package. Circuits in an MCM are normally horizontal.